

NEUROLOGICALLY MOTIVATED SELF-SUPERVISED LEARNING FOR BILATERAL GAIT GEOMETRY

[Authors and affiliations to be inserted in the RISEx template]

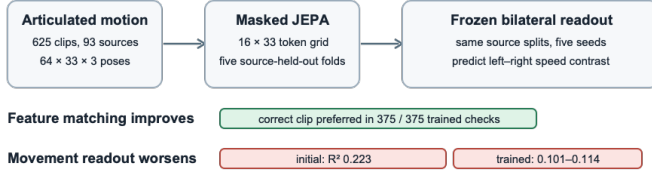


Figure 1: Study design and primary result: correctly matched latent features improve while the bilateral movement readout declines.

INTRODUCTION

Learning from videos of people could help robots anticipate human motion, but an internal feature match is an incomplete test of what a representation preserves. We evaluate a skeleton JEPA with a geometric relation that has a known answer. Reflection swaps left and right anatomy and reverses a signed speed difference. Gait asymmetry has been studied in stroke, Parkinson’s disease, cerebral palsy, and muscular disorders [1–3]. Those studies motivate the relation; this work neither diagnoses these conditions nor estimates disease severity.

MATERIALS AND METHODS

We selected 625 pose sequences from 93 GAVD source videos [4]. Processing is clip-local. It retains all 33 landmarks in a 64-by-33-by-3 array, then packs four frames per token to preserve a 16-by-33 spatial-temporal grid. The online encoder receives visible tokens, while a predictor estimates hidden teacher features. The teacher follows the online encoder by exponential moving average. This pretraining uses no movement target or disease label.

The target averages normalized left-minus-right median speed differences for five anatomical pairs. A frozen ridge readout is fitted only on outer-training sources. Five outer source-video folds keep test clips, transformed views, encoder training, and supervised readout fitting separate. Five seeds share initialization, source draws, views, and updates within each mask comparison. We report source-balanced scores and paired source resampling that keeps all clips and seed predictions from a source together.

RESULTS AND DISCUSSION

The initial encoder achieved $R^2 = 0.223$. Trained teachers achieved 0.101–0.114; every training arm was lower than its matched initialization (difference -0.109 to -0.122). Conversely, correct-clip targets had lower feature error in 375/375 trained diagnostic checks, versus 33/75 at initialization. Alternative motion and connected-region masks gave intervals spanning zero relative to matched random masks.

The result is a controlled warning for learned articulated-motion features. It remains conditional on the target and readout: preparation changes the score, and the larger readout combines motion, dimension, and missingness changes. It therefore cannot establish a general loss of geometric information or a robotics failure.

CONCLUSIONS

Latent feature correspondence and recovery of bilateral movement can disagree. A geometric observable, matched initialization, and source-held-out test split provide a compact evaluation pattern for AI representations of articulated motion.

REFERENCES

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- [3] Brændvik SM et al. Front Neurol 10:1399, 2020.
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